ITA03-MOBILE COMPUTING
ASSIGNMENT 3

ASSIGNMENT 3 – CO3

CO3 Coverage: Introduction to wireless networks – mobile IP - issues in mobile IP differences between cellular and ad hoc wireless networks- applications - routing protocols – Proactive & reactive Routing Protocols.

Mode: Pair Work – 2 Students per Group
Duration: 1 Week
Total Marks: 50

Pair-Work Principle

Both students are responsible for understanding the complete assignment.

However, each question contains clearly identified individual responsibilities so that the contribution and understanding of each student can be independently verified.

Student A and Student B must not divide the assignment question-wise. Both students must participate in every question.

The individual responsibilities intentionally alternate across questions so that both students demonstrate evidence of:

Concept Understanding → Network Analysis → Protocol Design → Implementation/Simulation → Evaluation

Q1 – Wireless Network Architecture and Mobility Analysis 6 Marks Part A –

Introduction to Wireless Networks

Shared Group Work

Together:

- Understand the characteristics and components of wireless networks. ● Identify mobile nodes, wireless access points/base stations and communication links.
- Develop a suitable wireless network architecture.
- Identify bandwidth, connectivity and mobility constraints.
- analyse communication between mobile users and network infrastructure.
- Implement or simulate a small wireless communication scenario and test the communication flow.

Student A – Network Architecture Analysis

Student A must individually:

- Draw and represent the wireless network architecture.
- Identify the major network components and their roles.
- analyse wireless communication characteristics.
- Analyse the effect of limited bandwidth and wireless connectivity. ● analyse how mobile nodes communicate through the wireless infrastructure.

Student B – Mobility and Performance Analysis

Student B must individually:

- Design representative mobility scenarios.
- Analyse handoff and connectivity interruption.
- Study packet loss and transmission delay.
- Analyse the effect of changing network conditions.
- Compare communication behaviour under different mobility conditions.

Part B – Wireless Mobility Scenario

Shared Group Work

Together, develop and analyse a scenario in which a mobile user moves between wireless coverage areas.

Student A – Analytical Investigation

Student A must:

- Define successful communication and interrupted communication cases. ● Identify the sequence of mobility events.
- Record communication delay and packet delivery behaviour.
- Investigate increasing mobility conditions.
- Present the analytical evidence.

Student B – Mobility Logic

Student B must:

- Formulate the mobility decision sequence.
- analyse when a mobile node changes its point of attachment. ● Identify which communication conditions are affected.
- analyse the termination or completion condition for the scenario. ● Relate repeated mobility to communication efficiency.

Individual Verification

Faculty may ask Student A to analyse the wireless architecture and Student B to analyse the mobility scenario. Therefore, primary responsibility does not mean exclusive knowledge.

Q2 – Mobile IP and Mobility Management 7 Marks Shared Group Work

The pair must:

- Study Mobile IP architecture and terminology.
- Identify Mobile Node, Home Agent, Foreign Agent and Correspondent Node. ● analyse agent discovery and registration.
- Analyse packet delivery to a mobile node.
- analyse tunnelling and mobility support.
- Conduct a small simulation or worked example of Mobile IP communication. **Student A –**

Mobile IP Operation

Student A is individually responsible for:

- analyseing the role of the Mobile Node.
- analyseing the role of the Home Agent.
- analyseing the role of the Foreign Agent.
- analyseing the role of the Correspondent Node.
- Describing registration and location updating.
- analyseing the packet-delivery and tunnelling process.

Student B – Mobile IP Challenge Analysis

Student B is individually responsible for:

- Identifying Mobile IP challenges.
- Analysing registration delay.
- Analysing tunnelling overhead.
- analyseing triangular routing.
- Analysing packet loss during mobility.
- interpreting security and scalability considerations.

Joint Design Question

Together answer:

If a mobile user frequently changes its network of attachment during an active communication session, what problems can occur in Mobile IP and how can they be reduced?

Evidence Expected

- Mobile IP architecture diagram.
- Registration/packet-flow sequence.
- Identification of mobility-related delays and overhead.
- Reasoned justification of the proposed improvement.

Cross-Understanding Requirement

Student A must be able to analyse the major Mobile IP challenges.

Student B must be able to analyse the complete Mobile IP registration and packet delivery process.

Both students must be able to defend the answer.

Q3 – Cellular Networks versus Ad Hoc Wireless Networks 7 Marks Shared Group Work

Together:

- Study cellular wireless network architecture.
- Study ad hoc wireless network architecture.
- Identify infrastructure-based and infrastructure-less communication. ● Compare mobility and topology characteristics.
- Compare routing requirements.
- Develop a technical comparison diagram and table.

Student A – Cellular Network Investigation

Student A must individually:

- Represent cellular network architecture.
- analyse the role of base stations and infrastructure.
- Analyse centralized communication.
- analyse handoff between cells.
- Analyse scalability and coverage.
- Identify suitable cellular applications.

Student B – Ad Hoc Network Investigation

Student B must individually:

- Represent ad hoc network architecture.
- analyse communication between mobile nodes.
- Analyse dynamic topology.
- analyse multi-hop communication.
- Analyse the absence of fixed infrastructure.
- Identify suitable ad hoc applications.

Comparative Analysis

Parameter	**Cellular Network**	**Ad Hoc Network**
Infrastructure		
Communication		

Routing		
Mobility		
Topology		
Scalability		
Handoff		
Failure Handling		
Applications		

Joint Engineering Decision

Together determine:

For an emergency/disaster environment where fixed infrastructure may be unavailable, which network architecture would you select and why?

Cross-Defence
Student A must analyse why cellular networks depend on infrastructure.

Student B must analyse how ad hoc networks continue communication without fixed infrastructure.

Q4 – Applications of Mobile IP and Routing Requirements 7 Marks Shared

Group Work

Together:

- Identify real-world applications requiring mobility support.
- Select one application scenario such as emergency communication, mobile healthcare or intelligent transportation.
- Identify the mobile entities and communication requirements.
- Apply Mobile IP concepts to the selected scenario.
- Identify the routing requirements created by mobility.
- Evaluate the suitability of the proposed solution.

Student B – Application Architecture

Student B must individually:

- Select and describe an appropriate Mobile IP application.
- Identify mobile nodes and communication endpoints.
- Design the application network architecture.
- analyse how Mobile IP supports user mobility.
- Identify required network components.
- analyse the communication flow.

Student A – Performance and Requirement Analysis

Student A must individually:

- Analyse communication delay.
- Analyse packet delivery and connectivity.
- Identify mobility requirements.
- Analyse interruption during movement.
- Identify security and reliability requirements.
- Evaluate Mobile IP suitability.

Q5 – Proactive Routing Protocols in Mobile Networks 6 Marks
Shared Group Work

Together:

- Study proactive routing protocols.
- Understand continuous routing-table maintenance.
- Select a suitable protocol such as DSDV.
- Develop a mobile-node topology.
- Analyse routing-table updates.
- Evaluate routing behaviour under mobility.

Student A – Protocol Design

Student A must individually:

- analyse the principle of proactive routing.
- analyse DSDV operation.
- analyse routing-table maintenance.
- analyse sequence numbers and route updates.
- Draw a sample routing topology.
- Demonstrate route information before and after a topology change. **Student B –**

Performance Investigation

Student B must individually:

- Analyse routing-control overhead.
- Analyse route availability.

- Study convergence behaviour.
- Analyse control-message transmission.
- Compare behaviour for different network sizes.
- Evaluate delay and packet delivery.

Q6 – Reactive Routing and Protocol Comparison 7 Marks Student A – Reactive Routing Investigation

Student A takes primary responsibility for reactive routing.

- analyse the concept of reactive routing.
- Study AODV.
- analyse Route Request (RREQ).
- analyse Route Reply (RREP).
- analyse Route Error (RERR).
- Analyse route discovery and route maintenance.
- analyse the effect of mobility on route discovery.

Student B – Comparative Verification

Student B must provide a comparative analysis of AODV and DSDV.

- Compare route discovery mechanisms.
- Compare routing-table requirements.
- Analyse control overhead.
- Compare initial communication delay.
- Analyse mobility handling.
- Identify suitable application scenarios.
- Justify the selection of a protocol for a given network condition. **Joint**

Question

For a highly dynamic mobile network with frequent topology changes, would you select DSDV or AODV? Justify your decision using routing overhead, delay and mobility considerations.

DAY-4 DESIGN CHALLENGE

Q7 – Design a Mobile Emergency Communication Network 10 Marks The Day-4 problem must be solved collaboratively.

Neither student may work on this question independently and simply combine the answers later.

Scenario

During a natural disaster, ambulances, police vehicles, rescue teams and medical personnel move continuously across different geographical areas. Fixed communication infrastructure may become unavailable, while mobile users must exchange critical information with minimum delay.

Stage 1 – Independent Design Proposal

Before interpreting the solution with each other:

Student A

Independently propose:

- A suitable wireless network architecture.
- A mobility-management strategy.
- The role of Mobile IP.
- A possible communication model.

Student B

Independently propose:

- A suitable routing strategy.
- Proactive/reactive protocol selection.
- Route maintenance approach.
- Failure-handling mechanism.
- Performance parameters.

Each student's initial proposal must be retained in the submission. **Stage 2 –**

Compare the Two Designs

The pair must compare where the designs agree, where they differ, strengths and weaknesses, Mobile IP requirements, routing suitability, expected overhead, expected delay and handling of node mobility.

Stage 3 – Joint Final Design

Using the strongest ideas from the interpretion, jointly develop the final mobile emergency communication solution.

- Network architecture showing mobile nodes, wireless links and routing infrastructure.
- Mobile IP components: Mobile Node, Home Agent, Foreign Agent and Correspondent Node.
- Registration and tunnelling process.
- Selected routing protocol and justification.
- Failure handling for node/link failure and network disconnection. ● Performance evaluation using delay, packet delivery and routing overhead.

Stage 4 – Divide the Validation Responsibilities

Student A – Analytical Validation

- Analyse Mobile IP overhead.
- Analyse mobility-related delay.
- Evaluate packet delivery.
- Analyse communication interruption.
- Justify the selected mobility-management mechanism.

Student B – Experimental Validation

- Design network test cases.
- Test different numbers of mobile nodes.
- Test different mobility conditions.
- Compare proactive and reactive routing behaviour.
- Record delay and packet-delivery results.
- Present experimental evidence.

Stage 5 – Cross-Defence

Student A must analyse: Why was the selected routing protocol appropriate for the emergency mobile network?

Student B must analyse: How does Mobile IP maintain communication when a mobile node changes its network location?

Stage 6 – Joint Evaluation

- Why the proposed network architecture is suitable.
- Why Mobile IP is required or not required.
- Why the selected routing protocol is appropriate.
- What network overhead is introduced.
- How mobility affects performance.
- What happens when connectivity is lost.
- Under what conditions the proposed solution may not be appropriate.

Conclude: What characteristics should a network designer analyse before selecting Mobile IP and a routing protocol for a highly mobile wireless environment?

CONTRIBUTION RECORD

Question	**Student A – Primary Contribution**	**Student B – Primary Contribution**	**Joint Work**
Q1	Network Architecture	Mobility Analysis	Wireless Network Analysis

Q2	Mobile IP Operation	Mobile IP Challenges	Mobile IP Architecture

Q3	Cellular Network	Ad Hoc Network	Comparative Analysis
Q4	Performance Analysis	Application Architecture	Application Evaluation
Q5	Protocol Design	Performance Investigation	Proactive Routing
Q6	Reactive Routing	DSDV–AODV Comparison	Routing Evaluation
Q7	Analytical Validation	Experimental Validation	Final Network Design

INDIVIDUAL VERIFICATION

Faculty may select any question, including a component for which the student was not designated as the primary contributor.

- analyse wireless network architecture.
- analyse Mobile IP operation and registration.
- analyse mobility challenges.
- Compare cellular and ad hoc wireless networks.
- analyse applications of Mobile IP.
- analyse proactive routing.
- analyse reactive routing.
- Compare DSDV and AODV.
- Justify protocol selection.
- Interpret experimental results.
- Justify network design decisions.
- Respond to a modified network condition.

A student should not answer: "My partner did this section."

Understanding of the complete submitted work is a requirement of the assignment. **NOTE:**

RECOMMENDED FACULTY VERIFICATION MODEL (VIVA) For each student,

conduct a brief 4–5 minute verification.

Question 1 – Primary responsibility

Ask one technical question from the component for which the student had primary responsibility.
Question 2 – Partner's responsibility

Ask one question from a component primarily completed by the partner. **Question 3**

– Transfer question

Modify a condition and ask the student to reason about the consequence.

- If a mobile node moves from one network to another during an active session, what happens to its IP connectivity and how does Mobile IP address the problem?
- If the number of mobile nodes increases significantly, how would AODV and DSDV differ in routing overhead?
- If the network topology changes very frequently, which routing approach would you select and why?
- If the fixed infrastructure becomes unavailable during a disaster, how would an ad hoc network support communication?

This verification establishes individual attainment without requiring a second complete individual assignment.
SIMATS ENGINEERING

MOBILE COMPUTING

STANDARD FOR ALGORITHMIC PSEUDOCODE

Applicable to: Mobile Computing Assignments and Exercises

1. What is Mobile Computing Pseudocode?

Mobile Computing pseudocode is a clear, precise and language-independent representation of a mobile networking algorithm. It describes mobility management, packet forwarding, route discovery, route maintenance, handoff, and routing decisions without depending on Java, Python, C, C++, Android, or simulator-specific syntax.

A good Mobile Computing algorithm must make the network state and communication logic visible. It should show what happens when a mobile node moves, when a route exists or fails, when a registration is required, and when a packet must be forwarded through a tunnel or discovered route.

Remember

- Pseudocode is not English prose.
- Pseudocode is not programming code.

- Pseudocode is an implementation-independent specification of mobile-network logic.
- Every important networking decision must be explicit.
- Abstraction is permitted; ambiguity is not.

2. Standard Structure

Use the following structure for a complete Mobile Computing algorithm.

ALGORITHM AlgorithmName(parameters)

INPUT:

Clearly describe nodes, packets, addresses, routes or network state OUTPUT:

Clearly describe the expected mobility/routing result

BEGIN
INITIALISE required variables/data structures

Detect or receive relevant network information

Execute mobility/routing decisions

UPDATE network state

RETURN result

END

Example

ALGORITHM CheckMobileNodeLocation(MN, HomeNetwork, CurrentNetwork) INPUT:

MN – mobile node

HomeNetwork – node's registered home network

CurrentNetwork – network where MN is currently attached

OUTPUT:

Mobility status

BEGIN

IF CurrentNetwork = HomeNetwork THEN

status ← HOME_NETWORK

ELSE

status ← VISITED_NETWORK

END IF

RETURN status

END

3. Standard Keywords and Notation

PURPOSE	**USE**
Declare algorithm	ALGORITHM

Specify input	INPUT:
Specify output	OUTPUT:
Start algorithm	BEGIN
End algorithm	END
Assignment	←
Condition	IF ... THEN
Alternative	ELSE
Multiple conditions	ELSE IF
Loop	FOR ... DO / WHILE ... DO
Return result	RETURN
Boolean operations	AND, OR, NOT
Packet operation	FORWARD / DELIVER / DROP
Route operation	ADD ROUTE / REMOVE ROUTE / UPDATE ROUTE

Message operation	SEND / RECEIVE
Timer	START TIMER / EXPIRE TIMER

Use explicit state names such as HomeAddress, CareOfAddress, NextHop, SequenceNumber, HopCount, RouteState, LinkState and RegistrationStatus.

4. Wireless Network Representation

Represent a mobile network as a graph G = (V, E), where V is the set of mobile nodes and infrastructure nodes, and E represents currently usable communication links.

NETWORK G = (V, E)

V = {mobile nodes, base stations, access points, gateways}

E = {wireless links currently available}

link(u,v) = communication link between u and v

cost(u,v) = communication cost of link (u,v)
status(u,v) ∈ {UP, DOWN}

Network State

STATE:

Location[MN] = current network/location of mobile node MN

NextHop[D] = next hop toward destination D

RouteState[D] ∈ {VALID, INVALID, UNKNOWN}

SequenceNumber[D] = latest known destination sequence number

HopCount[D] = number of hops in current route

LastUpdate[D] = time of most recent route information

5. Wireless Network Discovery

Wireless nodes must determine whether they are connected to a suitable network and whether neighboring nodes are reachable. The discovery procedure is especially important in dynamic environments.

ALGORITHM DiscoverWirelessNeighbors(MN, TransmissionRange) INPUT:

MN – mobile node

TransmissionRange – maximum discovery range

OUTPUT:

NeighborSet

BEGIN

NeighborSet ← EMPTY_SET

BROADCAST HELLO from MN

WAIT for neighbor responses

FOR each Response received DO

IF SignalStrength(Response) > MIN_SIGNAL THEN

IF Distance(MN, Response.Node) ≤ TransmissionRange THEN
ADD Response.Node to NeighborSet

END IF

END IF

END FOR

RETURN NeighborSet

END

6. Mobile IP – Basic Operation

Mobile IP allows a mobile node to retain a permanent home address while changing its point of network attachment. The algorithm must maintain a binding between the permanent home address and the current care-of address.

Mobile IP Entities

- Mobile Node (MN)
- Home Network
- Home Agent (HA)
- Foreign Network / Visited Network
- Foreign Agent (FA) in traditional Mobile IPv4 terminology
- Care-of Address (CoA)
- Correspondent Node (CN)

ALGORITHM MobileIPLocationUpdate(MN, HA, CurrentNetwork)

INPUT:

MN – mobile node

HA – home agent

CurrentNetwork – visited network

OUTPUT:

Registered care-of address

BEGIN

IF CurrentNetwork = MN.HomeNetwork THEN

CoA ← MN.HomeAddress
RegistrationStatus ← HOME

ELSE

CoA ← ObtainCareOfAddress(CurrentNetwork) SEND

RegistrationRequest(MN.HomeAddress, CoA) TO HA WAIT FOR

RegistrationReply

IF RegistrationReply = ACCEPTED THEN

RegistrationStatus ← REGISTERED

ELSE

RegistrationStatus ← FAILED

END IF

END IF

RETURN CoA, RegistrationStatus

END

7. Mobile IP Agent Discovery

ALGORITHM MobileIPAgentDiscovery(MN)

INPUT:

MN – mobile node

OUTPUT:

Detected mobility agent or network information BEGIN

LISTEN for AgentAdvertisement

IF Advertisement received THEN

Extract NetworkAddress

Extract AgentType

IF AgentType = HOME_AGENT THEN
Set HomeAgentAddress

ELSE IF AgentType = FOREIGN_AGENT THEN Set

ForeignAgentAddress

END IF

RETURN AgentType, NetworkAddress ELSE

SEND AgentSolicitation

WAIT for Advertisement

IF Advertisement received THEN

RETURN detected agent information

ELSE

RETURN AGENT_NOT_FOUND

END IF

END IF

END

8. Mobile IP Registration

ALGORITHM RegisterMobileNode(MN, HA, CoA)

INPUT:

MN – mobile node

HA – home agent

CoA – current care-of address

OUTPUT:

Registration result

BEGIN

Create RegistrationRequest
Request.HomeAddress ← MN.HomeAddress

Request.CareOfAddress ← CoA

Request.Lifetime ← RequestedLifetime

Request.Identifier ← GenerateRegistrationIdentifier()

Authenticate Request

SEND Request TO HA

WAIT FOR RegistrationReply

IF Reply.Status = ACCEPTED THEN

Binding[MN.HomeAddress] ← CoA

RegistrationExpiry ← CurrentTime + Reply.Lifetime

RETURN REGISTRATION_SUCCESS

ELSE

RETURN REGISTRATION_FAILURE

END IF

END

9. Mobile IP Packet Tunnelling

The home agent can forward packets toward the mobile node's current location by tunnelling them to the registered care-of address.

ALGORITHM TunnelPacketToMobileNode(Packet, DestinationHomeAddress) INPUT:

Packet – original IP packet

DestinationHomeAddress – mobile node home address

OUTPUT:

Packet delivered through tunnel or DROP

BEGIN
IF Binding[DestinationHomeAddress] does not exist THEN

DROP Packet

RETURN DELIVERY_FAILURE

END IF

CoA ← Binding[DestinationHomeAddress]

EncapsulatedPacket ← ENCAPSULATE(Packet, CoA)

FORWARD EncapsulatedPacket TO CoA

IF TunnelDeliverySuccessful THEN

Decapsulate at tunnel endpoint

DELIVER original Packet TO MobileNode

RETURN DELIVERY_SUCCESS

ELSE

RETURN DELIVERY_FAILURE

END IF

END

10. Mobile IP Challenges

CHALLENGE	**ALGORITHMIC IMPLICATION**
Handoff delay	Detect movement and update location quickly
Registration overhead	Limit unnecessary signalling
Packet loss	Buffer, retransmit or reroute when appropriate
Triangular routing	Evaluate indirect path and forwarding delay
Security	Authenticate registration and protect mobility messages
Scalability	Avoid excessive control traffic

Wireless errors	Detect link failure and retry or select alternate path
Changing bandwidth	Adapt routing and transmission decisions

Challenge Detection Algorithm

ALGORITHM DetectMobileIPChallenge(MN, Route, Registration) INPUT:

MN, current route, registration state

OUTPUT:

Identified challenge

BEGIN

IF RegistrationStatus = FAILED THEN

RETURN REGISTRATION_PROBLEM

ELSE IF HandoffInProgress = TRUE THEN

RETURN HANDOFF_DELAY

ELSE IF PacketLossRate > MAX_LOSS THEN

RETURN PACKET_LOSS

ELSE IF RouteDelay > MAX_DELAY THEN

RETURN ROUTING_DELAY

ELSE IF SecurityValidation = FAILED THEN

RETURN SECURITY_PROBLEM

ELSE

RETURN NO_MAJOR_PROBLEM

END IF

END

11. Handoff Management

ALGORITHM PerformHandoff(MN, OldNetwork, NewNetwork)

INPUT:

MN – mobile node

OldNetwork – previous point of attachment

NewNetwork – target network

OUTPUT:

Handoff status

BEGIN

Detect NewNetwork

Measure signal quality

IF SignalQuality(NewNetwork) < MIN_SIGNAL THEN

RETURN HANDOFF_REJECTED

END IF

Obtain new network address or care-of address Pause non-

critical transmissions if required

UPDATE mobility registration

UPDATE routing information

IF RegistrationStatus = SUCCESS THEN

SWITCH active attachment TO NewNetwork

Resume data transmission

RETURN HANDOFF_SUCCESS

ELSE

Restore OldNetwork if available

RETURN HANDOFF_FAILURE

END IF
END

12. Cellular vs Ad Hoc Wireless Networks

PARAMETER	CELLULAR NETWORK	AD HOC NETWORK

Infrastructure	Base stations / access infrastructure	No fixed infrastructure required
Topology	Planned cells	Dynamic topology
Routing	Infrastructure-assisted	Distributed multi-hop routing
Node role	Mobile device mainly communicates through infrastructure	A node can act as host and router
Mobility support	Cellular mobility management	Distributed route discovery/maintenance
Deployment	Requires pre-existing infrastructure	Can be rapidly deployed
Typical environment	Public cellular service	Emergency, military, temporary or field networks
Failure behaviour	Infrastructure failure can affect service	Alternate peer routes may be possible

13. Selecting Cellular or Ad Hoc Architecture

ALGORITHM SelectWirelessArchitecture(Scenario)

INPUT:

InfrastructureAvailability

NodeMobility

DeploymentTime

CoverageRequirement

MultiHopRequirement

OUTPUT:

Recommended architecture
BEGIN

IF InfrastructureAvailability = AVAILABLE AND

CoverageRequirement = LARGE THEN

Architecture ← CELLULAR

ELSE IF DeploymentTime = IMMEDIATE AND

InfrastructureAvailability = UNAVAILABLE THEN

Architecture ← AD_HOC

ELSE IF MultiHopRequirement = REQUIRED THEN

Architecture ← AD_HOC

ELSE

Architecture ← CELLULAR

END IF

RETURN Architecture

END

14. Applications of Mobile IP

- Mobile employees accessing organizational services while moving between networks.
- Continuous Internet access across different points of attachment. ● Emergency response and disaster recovery communication.
- Mobile healthcare and ambulance communication.
- Transportation and fleet monitoring.
- Field engineering and remote maintenance.
- Temporary communication networks.
- Location-independent access to network applications.

ALGORITHM SelectMobileIPApplication(Requirement)

INPUT:

Requirement – mobility and communication requirement

OUTPUT:
Suitable application category

BEGIN

IF Requirement requires continuous Internet access while moving THEN RETURN

MOBILE_INTERNET_ACCESS

ELSE IF Requirement involves emergency vehicle mobility THEN RETURN

EMERGENCY_COMMUNICATION

ELSE IF Requirement requires remote patient/device connectivity THEN RETURN

MOBILE_HEALTHCARE

ELSE IF Requirement involves moving vehicles or fleets THEN

RETURN TRANSPORTATION

ELSE

RETURN GENERAL_MOBILE_ACCESS

END IF

END

15. Routing Protocols in Mobile Networks

Routing protocols determine how packets travel through changing mobile topologies. For this unit, routing is studied using the two major approaches: proactive and reactive.

Routing State

STATE:

RouteTable[Destination] = {NextHop, HopCount, SequenceNumber, State} State ∈

{VALID, INVALID, UNKNOWN}

NeighborTable = currently reachable neighbors

RouteRequestID = identifier for route discovery

RouteLifetime = validity period

16. Proactive Routing

A proactive protocol maintains route information before a packet transmission request occurs. This can reduce route-discovery delay but may create periodic control overhead.

ALGORITHM ProactiveRouteMaintenance(Node)

INPUT:

Node – mobile node

OUTPUT:

Updated routing table

BEGIN

INITIALISE RouteTable

PERIODICALLY DO

Broadcast current routing information

RECEIVE routing updates from neighbors

FOR each Update DO

Validate Update

FOR each Destination in Update DO

Calculate NewMetric

IF NewMetric < CurrentMetric(Destination) OR

Update.SequenceNumber > CurrentSequenceNumber(Destination) THEN UPDATE

RouteTable[Destination]

Set NextHop and HopCount

END IF

END FOR

END FOR

Remove expired or invalid entries

END PERIODICALLY
RETURN RouteTable

END

17. Proactive Routing – DSDV-Style Sequence-Number Update

A destination-sequenced distance-vector approach uses route metrics and sequence numbers to maintain fresh routes and reduce routing loops.

ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)

INPUT:

Node – receiving node

ReceivedRoute = {Destination, NextHop, HopCount, SequenceNumber} OUTPUT:

Updated routing table

BEGIN

Destination ← ReceivedRoute.Destination

NewHopCount ← ReceivedRoute.HopCount + 1

NewSequence ← ReceivedRoute.SequenceNumber

IF Destination not in RouteTable THEN

ADD ReceivedRoute with NextHop = Node.Neighbor

ELSE IF NewSequence > RouteTable[Destination].SequenceNumber THEN REPLACE

RouteTable[Destination]

ELSE IF NewSequence = RouteTable[Destination].SequenceNumber AND

NewHopCount < RouteTable[Destination].HopCount THEN

REPLACE RouteTable[Destination]

END IF

RETURN RouteTable

END

18. Reactive Routing

A reactive protocol discovers a route only when a source needs to send data and no valid route is available. Route discovery and route maintenance are therefore demand driven.

ALGORITHM ReactiveRouteDiscovery(Source, Destination)

INPUT:

Source – requesting node

Destination – required destination

OUTPUT:

Route or ROUTE_NOT_FOUND

BEGIN

IF RouteTable[Destination].State = VALID THEN

RETURN RouteTable[Destination]

END IF

RouteRequestID ← RouteRequestID + 1

Create RouteRequest(Source, Destination, RouteRequestID)

Broadcast RouteRequest to neighbors

WAIT for RouteReply or RouteFailure

IF RouteReply received THEN

Establish reverse path

Establish forward path

Mark RouteTable[Destination] as VALID

RETURN RouteTable[Destination]

ELSE

RETURN ROUTE_NOT_FOUND

END IF
END

19. Reactive Routing – AODV-Style Route Request

ALGORITHM ProcessRouteRequest(Node, Request) INPUT:

Node – current receiving node

Request = {Source, Destination, RequestID, HopCount}

OUTPUT:

Route discovery action

BEGIN

IF Request already processed THEN

DISCARD Request

RETURN

END IF

Record reverse route toward Request.Source Mark

Request as processed

IF Node = Request.Destination THEN

Create RouteReply

SEND RouteReply toward Source

RETURN

END IF

IF ValidRouteTo(Request.Destination) THEN Create

RouteReply using valid route

SEND RouteReply toward Source

ELSE

Request.HopCount ← Request.HopCount + 1
FOR each Neighbor in NeighborTable DO

FORWARD Request TO Neighbor

END FOR

END IF

END

20. Reactive Route Maintenance

ALGORITHM MaintainReactiveRoute(Source, Destination) INPUT:

Source and Destination

OUTPUT:

Route status

BEGIN

MONITOR active route

IF LinkFailureDetected THEN

Mark failed link as INVALID

IF AlternateValidRoute exists THEN

Select alternate route

RETURN ALTERNATE_ROUTE_USED

ELSE

Generate RouteError

SEND RouteError toward affected source

IF Source still requires communication THEN Initiate

new RouteDiscovery

ELSE

RETURN ROUTE_TERMINATED
END IF

END IF

END IF

RETURN ROUTE_ACTIVE

END

21. Proactive vs Reactive Routing

FEATURE	**PROACTIVE**	**REACTIVE**
Route availability	Maintained before data transmission	Obtained when required
Route discovery delay	Low	Present when route is unknown
Control overhead	Periodic/continuous	Demand-driven
Stale information	Possible	Route is generally discovered when needed
Frequent traffic	Often suitable	May repeatedly use discovered routes
Infrequent traffic	May waste control traffic	Often efficient
Topology changes	Routing tables updated periodically	Route repair/discovery triggered

Typical example	DSDV-style	AODV-style

22. Proactive or Reactive – Decision Algorithm

ALGORITHM SelectRoutingApproach(TrafficRate, MobilityRate, DelayRequirement, Bandwidth)

INPUT:

TrafficRate – frequency of data communication

MobilityRate – frequency of topology changes
DelayRequirement – acceptable route setup delay

Bandwidth – available bandwidth

OUTPUT:

Recommended routing approach

BEGIN

IF DelayRequirement = VERY_LOW AND TrafficRate = HIGH THEN

Recommendation ← PROACTIVE

ELSE IF TrafficRate = LOW AND Bandwidth = LIMITED THEN

Recommendation ← REACTIVE

ELSE IF MobilityRate = HIGH AND TrafficRate = INTERMITTENT THEN

Recommendation ← REACTIVE

ELSE IF TrafficRate = HIGH AND DelayRequirement = LOW THEN

Recommendation ← PROACTIVE

ELSE

Recommendation ← EVALUATE_BOTH

END IF

RETURN Recommendation

END

23. Route Selection and Metric Calculation

FUNCTION CalculateRouteMetric(HopCount, Delay, LossRate, EnergyCost) INPUT:

HopCount, Delay, LossRate, EnergyCost

OUTPUT:

RouteMetric

BEGIN
RouteMetric ←

W1 × HopCount +

W2 × Delay +

W3 × LossRate +

W4 × EnergyCost

RETURN RouteMetric

END

ALGORITHM SelectBestRoute(RouteSet) INPUT:

RouteSet – candidate routes

OUTPUT:

BestRoute

BEGIN

BestRoute ← NULL

BestMetric ← INFINITY

FOR each Route in RouteSet DO

Metric ← CalculateRouteMetric(Route.HopCount,

Route.Delay,

Route.LossRate,

Route.EnergyCost)

IF Metric < BestMetric THEN

BestMetric ← Metric

BestRoute ← Route

END IF

END FOR
RETURN BestRoute

END

24. Packet Forwarding in a Mobile Network ALGORITHM

ForwardMobilePacket(Node, Packet) INPUT:

Node – current forwarding node

Packet – data packet

OUTPUT:

Delivery, forwarding or drop decision

BEGIN

Destination ← Packet.Destination

IF Node = Destination THEN

DELIVER Packet

RETURN DELIVERED

END IF

IF ValidRouteTo(Destination) THEN

NextHop ← RouteTable[Destination].NextHop IF

LinkStatus(NextHop) = UP THEN

FORWARD Packet TO NextHop

RETURN FORWARDED

ELSE

Mark route as INVALID

RETURN ROUTE_FAILURE

END IF

ELSE
RETURN ROUTE_DISCOVERY_REQUIRED

END IF

END

25. Routing Loop Detection

ALGORITHM DetectRoutingLoop(Path)

INPUT:

Path – ordered sequence of nodes

OUTPUT:

TRUE if loop exists, otherwise FALSE

BEGIN

Visited ← EMPTY_SET

FOR each Node in Path DO

IF Node IN Visited THEN

RETURN TRUE

END IF

ADD Node TO Visited

END FOR

RETURN FALSE

END

26. Manual Trace / Desk Check

Mobile routing pseudocode must be manually traceable. Record the node, current route state, received message, decision, next hop, and resulting route table state.

STEP	**NODE**	**EVENT**	**DECISION**	**ROUTE STATE**
1	S	Needs D	No valid route	UNKNOWN
2	S	Broadcast RREQ	Forward to neighbors	DISCOVERY
3	A	Receives RREQ	Record reverse route	DISCOVERY

4	D	Receives RREQ	Generate RREP	VALID
5	S	Receives RREP	Install route S→A→D	VALID
6	S	Data packet	Forward to A	ACTIVE

27. What Will NOT Be Accepted as Mobile Routing Pseudocode? Too vague:

Start

Find route

Send packet

Handle movement

Display result

Stop

There is insufficient computational logic.

Programming code disguised as pseudocode:

for(int i=0; i<n; i++) {

if(route[i].valid) send(packet);

}

Acceptable algorithmic form:
FOR each Route in RouteTable DO

IF Route.State = VALID THEN

FORWARD Packet TO Route.NextHop

RETURN FORWARDED

END IF

END FOR

RETURN ROUTE_DISCOVERY_REQUIRED

28. Programming-Language Syntax Should Not Be Used

- Avoid Java, Python, C, C++, Android API calls, or simulator-specific commands unless actual implementation code is explicitly requested.
- Use SEND, RECEIVE, FORWARD, DELIVER, DROP, UPDATE, ADD ROUTE and REMOVE ROUTE.
- Use ← for assignment.
- Use explicit network conditions instead of vague statements such as 'handle mobility'.

29. Comments

// Detect whether the current link is still usable

IF LinkStatus(NextHop) = DOWN THEN

Mark Route INVALID

Initiate route repair

END IF

Comments may clarify a non-obvious networking decision, but comments cannot replace the routing logic. The supplied DAA standard similarly requires the pseudocode its elf to analyse how the result is obtained. □filecite□turn0file0□L271 ₋ L277□

30. Quality Checklist

- Inputs are clearly defined.
- Outputs are clearly defined.
- Mobile-network states have clear meanings.
- Addresses, routes and sequence information are initialised.
- Packet forwarding conditions are explicit.
- Registration and handoff conditions are explicit.
- Route discovery and route repair have clear termination.
- Link failure and unavailable-route cases are addressed.
- The algorithm can be manually traced.
- Programming-language-specific syntax is avoided.
- Another programmer could implement the algorithm without guessing the logic.

31. How Mobile Computing Pseudocode Will Be Assessed

CRITERION	WHAT IS EXPECTED
Algorithmic Correctness	The mobility or routing logic correctly solves the stated problem.
Completeness	Inputs, outputs, state, decisions, loops, packet actions and termination are sufficiently specified.

Precision	Conditions such as route validity, link state, registration status and sequence number are explicit.
Networking Understanding	The solution correctly reflects Mobile IP, cellular/ad hoc networking or routing concepts.
Appropriate Abstraction	Network operations are represented clearly without unnecessary programming syntax.
Traceability	A small network scenario can be manually executed using the pseudocode.
Technical Justification	The selected proactive/reactive or mobility mechanism is supported by network characteristics.

32. Final Test

Before submitting any Mobile Computing pseudocode, ask:

- Can another competent programmer implement my algorithm without asking what I meant?
- Can the same algorithm be implemented in Java, Python, C/C++ or a network simulator without changing its underlying logic?
- Does the algorithm explicitly handle mobility, routing state and failure conditions?
- Can I trace the algorithm using a small mobile-network example? **33. THE**

MOBILE COMPUTING DESIGN RULE

UNDERSTAND → MODEL NETWORK → DEFINE STATE → DESIGN ALGORITHM → WRITE PSEUDOCODE → TRACE → TEST → REFINE → IMPLEMENT

Do not write the program first and convert it into pseudocode afterwards. Design the mobile-network behaviour first. The program or simulator is the implementation of that design. This follows the source handbook's final design principle:

THINK → DESIGN → WRITE PSEUDOCODE → TRACE → REFINE → IMPLEMENT.

ASSIGNMENT 3 – ANSWERS

Q1 – Wireless Network Architecture and Mobility Analysis (6 Marks)

Part A – Wireless Network Architecture

A wireless network allows devices to exchange data using radio waves instead of physical cables. A typical infrastructure-based network contains mobile nodes, wireless access points or base stations, a backhaul network, a router/gateway and a server or Internet connection.

Component	**Role**
Mobile Node (MN)	A phone, laptop or vehicle that sends and receives data while moving.
Access Point / Base Station	Provides radio coverage and connects mobile nodes to the wired/backhaul network.
Wireless Link	Carries frames between a mobile node and the access point, or between peer nodes.
Router / Gateway	Forwards packets between the access network and other networks.
Application Server / Correspondent Node	The endpoint that provides or receives the requested data.

Architecture representation

PSEUDOCODE
Mobile Node A)) wireless link ((Access Point / Base Station
Access Point ---- Backhaul / Router ---- Server or Correspondent Node
Mobile Node B)) wireless link ((Access Point / Base Station

Wireless communication characteristics

- Bandwidth is shared among users; more active users can reduce the throughput available to each user.
- Radio signals are affected by distance, obstacles, interference, fading and shadowing.
- Connectivity can be interrupted when a node moves outside coverage or changes access points.
- Mobility may trigger handoff, packet loss, retransmissions and additional delay.
- Security requires authentication, encryption and protection against unauthorised access.

Part B – Worked mobility scenario

Assume a mobile user moves from Access Point A to Access Point B while a server sends 100 packets. The following values are illustrative desk-check values, not measured simulation results.

Condition	Packets sent	Packets delivered	Packet loss	Delivery ratio
Stationary near AP A	100	98	2 packets	98%
Moving during handoff	100	90	10 packets	90%
After handoff completes	100	97	3 packets	97%

Step 1 – Packet delivery ratio = (Packets delivered ÷ Packets sent) × 100.

Step 2 – During handoff: (90 ÷ 100) × 100 = 90%.

Step 3 – Packet loss = Packets sent − Packets delivered = 100 − 90 = 10 packets.

Step 4 – Compare conditions: delivery falls from 98% while stationary to 90% during handoff, then improves to 97% after attachment to AP B.

Mobility decision sequence

PSEUDOCODE
ALGORITHM ManageWirelessMobility(MN)
INPUT: Current access point, candidate access points, signal quality
OUTPUT: Active access point and connectivity status
BEGIN
Measure signal quality of the current and candidate access points
IF current signal quality is acceptable THEN
Continue communication through the current access point
ELSE
Select a reachable candidate with acceptable signal quality
Authenticate MN with the candidate access point
UPDATE active attachment to the candidate access point
Restore or update routes and resume packet transmission

END IF
Record delay, lost packets and delivery ratio
RETURN connectivity status
END

Network components and their functions

- A mobile node is the user device that sends or receives information while moving. Examples include smartphones, tablets, laptops and vehicle-mounted terminals.
- An access point or base station provides the radio connection between mobile devices and the wider network. A gateway or router forwards traffic between the wireless access network, the Internet and application servers.
- The backhaul carries traffic from access points or base stations to core network services. If the backhaul is congested or unavailable, a strong local radio signal alone cannot guarantee successful end-to-end communication.

Effect of bandwidth and interference

- Wireless bandwidth is a shared resource. When many users transmit at the same time, contention and scheduling reduce the capacity available to each device.
- Interference from nearby networks, electronic equipment and overlapping channels can increase frame errors and retransmissions. These retransmissions consume additional airtime and increase delay.
- Adaptive modulation, channel selection, power control and suitable access-point placement can improve performance. These measures should be selected according to the environment rather than assuming that one setting works everywhere.

Mobility, handoff and continuity

- When a mobile device moves away from its serving access point, signal strength and link quality may decrease. The device may then scan for another access point and perform authentication and reassociation.
- During this transition, packets may be delayed, buffered, retransmitted or lost. Applications such as voice calls and live video are especially sensitive to long interruptions.
- Handoff performance can be improved through overlapping coverage, timely neighbour discovery, suitable signal thresholds and buffering. A device should avoid switching repeatedly between access points when signal levels fluctuate near a boundary.

Performance measures for the mobility test

- Packet delivery ratio measures the percentage of transmitted packets that reach the destination. Packet loss percentage measures the packets that fail to arrive during the observation period.
- Average end-to-end delay is calculated by adding the delays of delivered packets and dividing by the number of delivered packets. Handoff interruption time should also be recorded because an average delay can hide a short but important outage.

- The illustrative result in this answer shows a reduction in delivery during handoff. A real experiment should repeat the test under the same traffic rate and record the number of packets sent, received and lost in each mobility condition.

Conclusion: A wireless network must balance coverage, bandwidth, handoff time and packet delivery. Monitoring signal quality and preparing the target access point can reduce interruptions during movement.

Q2 – Mobile IP and Mobility Management (7 Marks)

Mobile IP allows a Mobile Node (MN) to keep its permanent home address while visiting another network. A Home Agent (HA) stores the current binding between the home address and the Care-of Address (CoA). In traditional Mobile IPv4, a Foreign Agent (FA) may assist the visiting node. A Correspondent Node (CN) sends packets to the MN.

Entity	**Function**
Mobile Node (MN)	Device that moves between networks and communicates using its home address.
Home Agent (HA)	Maintains the home-address-to-CoA binding and tunnels packets to the current location.
Foreign Agent (FA)	In traditional Mobile IPv4, provides a visited-network point of attachment and may supply a CoA.
Correspondent Node (CN)	Peer that sends data to or receives data from the MN.
Care-of Address (CoA)	Address representing the MN's current visited-network location.

Mobile IP registration and packet delivery

1. The MN detects that it has entered a visited network using agent advertisements or solicitation.
2. The MN obtains a CoA from the visited network or FA.
3. The MN sends a registration request containing its home address and CoA to the HA.
4. The HA authenticates and accepts the request, then stores the binding for the registration lifetime.
5. The CN sends a packet to the MN's home address.
6. The HA intercepts the packet, encapsulates it and tunnels it to the CoA.
7. The tunnel endpoint decapsulates the packet and delivers it to the MN.

PSEUDOCODE
ALGORITHM MobileIPRegistrationAndDelivery(MN, HA, CN, VisitedNetwork)
INPUT: MN home address, visited network, packet from CN
OUTPUT: Registration status and packet delivery status

BEGIN
IF MN is on its home network THEN
Deliver packets using normal IP routing
ELSE
CoA ← ObtainCareOfAddress(VisitedNetwork)
SEND RegistrationRequest(HomeAddress, CoA) TO HA
WAIT FOR RegistrationReply
IF RegistrationReply = ACCEPTED THEN
Binding[HomeAddress] ← CoA
WHEN packet arrives for HomeAddress at HA DO
Encapsulate packet with tunnel destination CoA
FORWARD encapsulated packet to CoA
Decapsulate at tunnel endpoint
DELIVER original packet to MN
END WHEN
ELSE
Record registration failure and retry if permitted
END IF
END IF
RETURN registration and delivery status
END

Challenges and improvement

Challenge	**Effect**	**Possible improvement**
Registration delay	Communication may pause while the new location is registered.	Use timely agent discovery and efficient registration procedures.

Tunnelling overhead	Encapsulation adds headers and consumes bandwidth.	Reduce unnecessary tunnelling and use efficient tunnel management.
Triangular routing	CN-to-HA-to-MN path can be longer than a direct path.	Use route optimisation where supported and securely configured.
Packet loss during handoff	Packets may arrive at the old location or be lost during transition.	Use buffering, prompt binding updates and transport-layer recovery.
Security	Forged registration could redirect traffic.	Authenticate registration and protect mobility signalling.
Scalability	Many mobile nodes increase signalling and binding state.	Use efficient registration timers and scalable agent management.

Purpose of a permanent home address

- The Mobile Node retains its home address even when it connects through a visited network. Applications and correspondent devices can continue using this stable address instead of learning a new address every time the user moves.
- The Care-of Address identifies the mobile node's current point of attachment. The Home Agent maintains a binding between the home address and the registered Care-of Address.
- This separation between identity and current location is the main idea behind Mobile IP. It allows mobility management to operate without requiring every correspondent node to track each movement.

Agent discovery and registration

- The mobile node first determines whether it is connected to its home network or a foreign network. In traditional Mobile IPv4, agent advertisements and solicitations help the node discover mobility agents.
- When a Care-of Address is obtained, the mobile node sends a registration request through the visited network. The Home Agent authenticates the request and updates the mobility binding if the registration is valid.
- Registration lifetime controls how long the binding remains valid. The mobile node must refresh the registration before expiry; otherwise, the Home Agent may stop forwarding packets to the old Care-of Address.

Packet tunnelling and delivery

- A correspondent node sends a packet to the mobile node's home address. The Home Agent intercepts the packet and encapsulates it inside another packet addressed to the Care-of Address.
- The tunnel carries the encapsulated packet across the network to the visited location. The tunnel endpoint removes the outer header and delivers the original packet to the mobile node.

- Tunnelling supports reachability but adds header overhead and may create a non-optimal route. Route optimisation can reduce this detour where the required security and protocol support are available.

Problems caused by repeated movement

- Frequent movement can cause repeated network discovery and registration messages. If registration takes too long, packets may continue to be sent toward an old location or may be discarded during the transition.
- Tunnelling adds processing and header overhead. Triangular routing may also increase path length when packets travel through the Home Agent even though a shorter path to the visited network exists.
- Rapid movement can cause overlapping registrations, stale bindings and packet reordering. Authentication and replay protection are necessary so that an attacker cannot redirect traffic by submitting a false location update.

Methods to improve Mobile IP performance

- Fast agent discovery and a suitable registration procedure can reduce the time required to establish a new binding. Buffering packets during handoff can reduce loss while the new path is being configured.
- Route optimisation can shorten packet paths, while hierarchical mobility management can reduce the distance over which location updates must travel. These approaches introduce their own state-management and security requirements.
- The network should monitor registration delay, handoff interruption, packet loss, tunnel overhead and binding-update frequency. Improvements should be verified using repeatable tests rather than assumed from the design alone.

Conclusion: Mobile IP supports location-independent communication, but frequent movement increases registration signalling, delay and packet-loss risk. Faster discovery, secure registration, buffering and route optimisation can reduce these effects.

Q3 – Cellular Networks versus Ad Hoc Wireless Networks (7 Marks)

A cellular network uses planned cells and fixed base stations. An ad hoc network is formed by wireless nodes without depending on fixed infrastructure; nodes can forward packets for one another over multiple hops.

Parameter	**Cellular Network**	**Ad Hoc Network**
Infrastructure	Uses base stations, core network and operator-managed links.	No fixed infrastructure is required for basic operation.
Communication	Usually mobile node to base station, then through the network.	Direct node-to-node or multi-hop forwarding.
Routing	Infrastructure/core network assists packet forwarding.	Distributed routing protocol discovers and maintains paths.

Mobility	Uses cellular handoff and mobility management.	Topology and routes change as nodes move.
Topology	Planned cells and managed coverage.	Dynamic topology determined by reachable neighbours.
Scalability	Can provide broad coverage with planned capacity expansion.	Routing and interference can become difficult as nodes increase.
Handoff	Managed transfer between cells/base stations.	Routes are repaired or rediscovered after links change.
Failure handling	Base-station/core failure can affect service in an area.	Alternate peer paths may keep communication working.
Applications	Public mobile service, broadband, fleet connectivity.	Disaster response, temporary events, military/field networks.

Simple architecture diagrams

PSEUDOCODE
CELLULAR: Mobile Node)) Base Station ---- Core Network ---- Server
AD HOC: Node A)) Node B)) Node C)) Node D
In an ad hoc path, intermediate nodes forward packets toward the destination.

Engineering decision for disaster conditions

If fixed infrastructure is unavailable, select an ad hoc wireless network as the immediate communication layer. Rescue teams, ambulances and police vehicles can exchange messages directly or through multi-hop relays. If a surviving cellular network is available, it can be used as an additional path rather than the only path.

Advantages: rapid deployment, no mandatory base station, flexible topology and possible alternate routes. Limitations: changing routes, limited radio range, interference, battery consumption and lower predictable capacity.

Cellular network architecture

- A cellular network divides a geographical area into cells. Each cell is served by a base station that manages radio communication with mobile devices in its coverage area.
- Base stations connect to a core network that performs mobility management, authentication, call or session control and access to external networks. This organised structure supports large numbers of users and planned frequency reuse.

- As a user moves between cells, the network coordinates handoff so that an ongoing call or data session can continue. The network operator can manage capacity, coverage and security centrally.

Ad hoc network architecture

- An ad hoc wireless network is formed by participating nodes without requiring a permanent base station. Each node can act as both an end device and a packet-forwarding router.
- If the destination is outside direct radio range, packets can travel through intermediate nodes. The route therefore depends on which neighbouring devices are currently reachable.
- Because nodes move and links change, the topology can change rapidly. Routing protocols must detect broken paths, discover alternatives and avoid forwarding loops.

Technical comparison of both architectures

- Cellular networks offer managed coverage, centralised authentication and predictable service planning when infrastructure is functioning. Their main dependency is the availability of base stations, backhaul and core-network services.
- Ad hoc networks can be deployed quickly and can continue local communication when fixed infrastructure is unavailable. Their limitations include variable capacity, changing routes, interference, limited radio range and battery consumption.
- Cellular communication usually relies on infrastructure-based paths, while ad hoc communication can use direct or multi-hop node-to-node paths. The choice depends on coverage, urgency, scale, reliability and available equipment.

Disaster-response network selection

- When base stations and backhaul are unavailable, an ad hoc network is a practical first communication layer for rescue teams operating in the affected area.
- Nearby ambulances, police vehicles and rescue teams can forward messages for one another. Nodes positioned at suitable locations may act as temporary relays to extend coverage.
- If any cellular, satellite or fixed network connection survives, it can be connected as an optional gateway. A hybrid design provides local resilience while retaining access to remote hospitals and command centres when an external path is available.

Conclusion: Cellular networks are suitable when managed infrastructure and wide-area coverage are available. Ad hoc networks are more suitable for rapidly deployed communication when infrastructure is damaged or absent.

Q4 – Applications of Mobile IP and Routing Requirements (7 Marks)

Selected application: Emergency ambulance and medical communication

An ambulance sends patient status, location and estimated arrival time to a hospital while travelling across different network areas. Mobile IP can preserve the ambulance device's home-address identity while it changes its point of attachment. Routing must provide reliable delivery with low delay.

Network entity	**Purpose**
Ambulance Mobile Node	Sends patient data, GPS location and status updates.
Home Agent	Maintains the ambulance node's home-address binding.

Visited network / Foreign Agent	Provides connectivity in the current coverage area and may provide the CoA in Mobile IPv4.
Hospital server / Correspondent Node	Receives the ambulance information and sends instructions.
Routing layer	Finds a valid path and repairs/discovers routes when links change.

Communication flow

1. The ambulance device connects to the current wireless network.
2. If it moves to a visited network, it obtains a CoA and registers with the HA.
3. The device sends a patient update to the hospital server.
4. When the hospital sends a response to the home address, the HA tunnels it to the registered CoA.
5. The routing layer forwards packets over the available links and monitors failures.
6. After handoff, the device updates mobility state and resumes communication.

Worked performance example

Assume 200 update packets are sent during a test. These are illustrative values for demonstrating the calculation only.

Step 1 – If 190 packets arrive, packet delivery ratio = (190 ÷ 200) × 100 = 95%.

Step 2 – Packet loss = 200 − 190 = 10 packets; packet loss percentage = (10 ÷ 200) × 100 = 5%.

Step 3 – If total end-to-end delay for 190 delivered packets is 3.8 seconds, average delay = 3.8 ÷ 190 = 0.02 seconds = 20 ms.

Metric	**Formula**	**Illustrative result**
Packet delivery ratio	Delivered ÷ Sent × 100	190 ÷ 200 × 100 = 95%
Packet loss	Sent − Delivered	200 − 190 = 10 packets (5%)
Average delay	Total delay ÷ delivered packets	3.8 s ÷ 190 = 0.02 s = 20 ms

Requirements and suitability

- Low delay for urgent patient status and hospital instructions.
- High packet delivery ratio and recovery from temporary link failures.
- Secure authentication and encryption for patient information.
- Continuous mobility management and prompt location updates.
- Backup communication path where possible, because Mobile IP alone does not guarantee connectivity if all links fail.

Mobile entities and information flow

- The ambulance contains a mobile node that sends patient condition, vehicle location and estimated arrival time. The hospital server or emergency coordination centre acts as the communication endpoint.
- The mobile node may move between cellular coverage areas or connect through a temporary wireless network. Mobile IP can preserve the node's home-address identity when the required mobility agents and network connectivity are available.
- Routing forwards packets across the current network path. Mobility management updates the node's location, while the application decides which information must be sent urgently and which information can be queued.

Quality-of-service requirements

- Critical alerts require low delay and high delivery probability. Routine telemetry, such as periodic status updates, can tolerate a longer delay if the latest information is eventually delivered.
- Packet priority should be assigned according to clinical importance. Emergency alarms and requests for immediate assistance should be transmitted ahead of non-urgent logs or background synchronisation.
- The design should record delay, packet delivery ratio, packet loss, jitter and service interruption. These measures help determine whether the network meets the requirements of the application.

Security and privacy requirements

- Patient information must be protected from unauthorised reading and modification. Authentication should confirm that the ambulance device and hospital endpoint are permitted to exchange data.
- Encryption protects information while it travels across wireless links. Access control and audit records help limit who can view or update patient details.
- Mobility registration must also be protected. A forged location update could redirect packets, so registration messages require suitable authentication and replay protection.

Reliability during temporary disconnection

- A mobile network may temporarily lose coverage while the ambulance travels through a weak-signal area. The application should store important updates locally and retry transmission when a usable link returns.
- Messages should include timestamps, identifiers and delivery status so that duplicate transmissions can be detected and older updates do not overwrite newer information.
- Mobile IP cannot deliver packets when every path to the mobile node is unavailable. A backup radio, alternate network or store-and-forward method is therefore needed for dependable emergency operation.

Conclusion: Mobile IP is useful when a mobile endpoint must remain reachable across networks. It should be combined with reliable routing, security and a backup communication method for emergency healthcare.

Q5 – Proactive Routing Protocols in Mobile Networks (6 Marks)

Proactive routing maintains routes before a data packet needs to be sent. Destination-Sequenced Distance-Vector (DSDV) is a proactive protocol that uses destination sequence numbers to identify fresh routes and reduce routing loops. Nodes periodically exchange routing information.

DSDV route update procedure

PSEUDOCODE
ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)
INPUT: Destination, advertised hop count, sequence number, neighbour
OUTPUT: Updated routing table
BEGIN
Destination ← ReceivedRoute.Destination
NewHopCount ← ReceivedRoute.HopCount + 1
IF Destination is not in RouteTable THEN
ADD route using neighbour as NextHop
ELSE IF ReceivedRoute.SequenceNumber > RouteTable[Destination].SequenceNumber THEN
REPLACE route with the newer sequence number
ELSE IF sequence numbers are equal AND NewHopCount < current hop count THEN
REPLACE route with the shorter route
ELSE
KEEP current route
END IF
RETURN RouteTable
END

Example topology and route-table update

Consider S–A–D as the initial path. If the link A–D fails and S learns a valid route S–B–C–D, the updated route uses the new next hop and hop count after receiving fresh sequence-number information.

Destination	**Before change: next hop / hops**	**After change: next hop / hops**	**Decision**
A	A / 1	A / 1	Keep valid route
D	A / 2	B / 3	Use fresh valid route through B–C

C	A / not direct	B / 2	Install route if update is newer

Performance analysis

Feature	**Effect in DSDV**
Route availability	Routes are normally ready before data transmission.
Initial delay	Low when a valid route is current.
Control overhead	Periodic updates consume bandwidth even when no application data is sent.
Topology change	Updates must propagate; convergence may take time.
Sequence numbers	Help select fresh routes and avoid routing loops.
Suitable conditions	Networks with regular traffic and where low route-setup delay is important.

Working principle of DSDV

- DSDV is a table-driven proactive routing protocol based on the distance-vector approach. Each node maintains routing information for reachable destinations, including the next hop, hop count and destination sequence number.
- Nodes exchange routing updates periodically and may also send triggered updates when important topology changes occur. Because routes are maintained in advance, a source can often forward a packet without waiting for a separate route-discovery process.
- Destination sequence numbers help nodes distinguish newer route information from stale information. A route with a newer sequence number is preferred, helping reduce the possibility of routing loops.

Example of route-table maintenance

- Assume the initial path from source S to destination D is S–A–D. The routing table at S records A as the next hop and a hop count of two.
- If link A–D fails, node A detects the failure and advertises that the route is no longer valid. S must not continue forwarding packets along the broken path.
- If an alternate path S–B–C–D is available, updated routing information can establish B as the next hop with a new hop count. The sequence number ensures that the newer route state is recognised.

Control overhead and convergence

- Periodic updates consume bandwidth even when no application data is being sent. As the number of nodes and destinations increases, routing tables and update messages can become larger.

- After a topology change, nodes need time to receive and process the new information. During convergence, some nodes may temporarily hold stale routes or drop packets until the updated information spreads.
- Update intervals create a trade-off. Frequent updates improve route freshness but increase control traffic, while infrequent updates reduce overhead but may delay detection of a broken path.

When proactive routing is suitable

- DSDV is useful when communication is frequent and low route-setup delay is important. It is more attractive when the topology changes at a moderate rate and the network can afford periodic control messages.
- For sparse or intermittent traffic, maintaining routes to many unused destinations may waste bandwidth and energy. In that situation, a reactive protocol may be more efficient.
- A fair evaluation should compare packet delivery ratio, average delay, route convergence time, control packets and routing overhead under the same mobility pattern and traffic load.

Conclusion: DSDV provides quick access to known routes, but periodic table updates can create overhead in highly dynamic networks. Route freshness and convergence should be evaluated as node mobility increases.

Q6 – Reactive Routing and Protocol Comparison (7 Marks)

Reactive routing discovers a route only when a source needs to send data and no valid route exists. Ad hoc On-Demand Distance Vector (AODV) uses Route Request (RREQ), Route Reply (RREP) and Route Error (RERR) messages to discover and maintain routes.

AODV route discovery

PSEUDOCODE
ALGORITHM AODVRouteDiscovery(Source, Destination)
INPUT: Source, Destination, neighbour table and route table
OUTPUT: Valid route or ROUTE_NOT_FOUND
BEGIN
IF a valid route to Destination exists THEN
RETURN existing route
END IF
Create a new RREQ with unique request identifier
Broadcast RREQ to neighbours
Each receiving node discards duplicate requests and records a reverse route to Source
IF Destination receives RREQ OR a permitted valid route is found THEN
Generate RREP and send it along the reverse path

Intermediate nodes install the forward route
Source marks the route VALID
RETURN route
ELSE
If discovery timeout expires, return ROUTE_NOT_FOUND
END IF
END

Route maintenance

1. Monitor active links used by the route.
2. When a link failure is detected, mark the affected route invalid.
3. If a valid alternate route exists, switch to it.
4. Otherwise, send an RERR message to affected source nodes.
5. If communication is still required, initiate a new route discovery.

DSDV versus AODV

Feature	**DSDV (Proactive)**	**AODV (Reactive)**
Route discovery	Routes maintained before a request.	Route discovered when needed.
Initial packet delay	Usually lower if route is current.	Higher when discovery is required.
Control overhead	Periodic updates even without data.	Mostly on-demand discovery and maintenance messages.
Mobility	Table updates must converge after changes.	Broken active routes trigger error handling and rediscovery.
Memory/state	Maintains routing information for destinations.	Maintains active/discovered route state and timers.
Best fit	Regular traffic and strict route-setup delay.	Intermittent traffic or limited bandwidth where unused routes should not be continually updated.

Decision for a highly dynamic network

For a highly dynamic network with frequent topology changes and intermittent communication, AODV is a reasonable choice because it discovers routes on demand and avoids continuous periodic updates for every destination. However, route discovery can add delay when communication starts, and frequent breakages can still cause RREQ/RERR overhead.

AODV route discovery process

- AODV is a reactive, on-demand routing protocol. When a source has data for a destination and no valid route is available, it broadcasts a Route Request message to its neighbours.
- Each receiving node records reverse-path information toward the source and checks whether it is the destination or has a sufficiently fresh route. If so, it sends a Route Reply back toward the source.
- The source uses the reply to establish a forward route. Sequence numbers help ensure that the selected route is based on sufficiently fresh information.

Purpose of RREQ, RREP and RERR

- RREQ initiates route discovery and carries information such as the source, destination and request identifier. Duplicate requests are discarded to limit unnecessary rebroadcasting.
- RREP provides a route back to the requesting source. Intermediate nodes can use the reply to establish next-hop information for the destination.
- RERR reports that a destination or link is no longer reachable through a previously active route. The source can then use an alternate route or start a new route discovery.

Route maintenance under mobility

- Nodes monitor the links used by active routes. A link failure may be detected through missing acknowledgements, neighbour information or link-layer notifications.
- When a route breaks, the affected route entry is marked invalid. If a valid alternate path exists, the node can use it; otherwise, an RERR message informs the source that route repair or rediscovery is required.
- Frequent movement can cause repeated discoveries and route errors. This increases control traffic and may interrupt application data, so mobility speed and node density must be considered when selecting AODV.

DSDV and AODV comparison

- DSDV maintains routes continuously and can provide a route immediately when data is ready. Its cost is periodic routing-table maintenance, including updates for destinations that may not currently carry traffic.
- AODV discovers routes only when needed and can reduce routine update traffic for intermittent communication. Its disadvantage is the initial route-discovery delay and additional control messages when active links fail.
- For a highly dynamic network, neither protocol is universally best. The selection should consider traffic frequency, mobility, bandwidth, route-setup delay, packet delivery and the energy available to mobile nodes.

Recommended evaluation method

- Use the same network area, radio range, traffic pattern, packet size and mobility model when comparing the protocols. Change one factor at a time, such as node count or movement speed.

- Record average end-to-end delay, packet delivery ratio, packet loss, routing-control packets and route-recovery time. Repeat each test with multiple random seeds to reduce the effect of one unusual run.
- Present the measured values separately from assumed examples. A protocol should be recommended only when the measured results support the stated application requirements.

Conclusion: Choose AODV when traffic is intermittent and bandwidth is limited; consider DSDV when routes must be available immediately and periodic control overhead is acceptable. The final selection should be validated using delay, packet delivery ratio and routing overhead under the expected mobility.

Q7 – Design a Mobile Emergency Communication Network (10 Marks)

Stage 1 – Independent design proposals

Student A – Architecture and mobility proposal

- Use a rapidly deployable ad hoc wireless network among ambulances, police vehicles, rescue teams and medical staff.
- Use surviving cellular or satellite/backhaul connectivity as an optional gateway when available.
- Use Mobile IP for mobile endpoints that need to remain reachable through a stable home address when a home/visited network and mobility agents are available.
- Use store-and-forward or local message buffering during temporary disconnection.

Student B – Routing proposal

- Use AODV for the changing ad hoc topology because it discovers routes on demand.
- Monitor link status and use RERR to notify sources when a route breaks.
- Use an alternate valid route if available; otherwise buffer critical messages and retry discovery.
- Measure end-to-end delay, packet delivery ratio, routing-control overhead and route-recovery time.

Stage 2 – Comparison of proposals

Design issue	**Student A proposal**	**Student B proposal**	**Joint decision**
Network architecture	Ad hoc nodes plus optional gateway	Routing across changing peer links	Use ad hoc mesh with optional surviving infrastructure gateway.
Mobility	Mobile IP where network-layer reachability is required	Route discovery/repair for peer paths	Mobile IP handles endpoint location; AODV handles ad hoc forwarding.
Failure handling	Buffer data during disconnection	Alternate route, RERR and rediscovery	Combine buffering, route repair and priority for critical messages.

Evaluation	Handoff and packet interruption	Delay, delivery and overhead	Test both mobility and routing metrics.

Stage 3 – Joint final architecture

PSEUDOCODE
Ambulance MN ---- wireless ---- Rescue Node ---- wireless ---- Hospital Gateway
| | |
Police Node ------ Rescue Team Node ------ Medical Node
Optional path: Mobile Node → visited network/FA → HA tunnel → Correspondent Node
Ad hoc links provide local multi-hop delivery when fixed infrastructure is unavailable.

Mobile IP components: MN is the mobile emergency device; HA maintains the home-address binding; FA is used in traditional Mobile IPv4 visited networks; CoA represents the current location; CN is the hospital server or other communication endpoint. In a purely disconnected ad hoc network with no reachable HA or external path, Mobile IP cannot itself create connectivity; local ad hoc routing and buffering are needed.

Joint route and failure algorithm

PSEUDOCODE
ALGORITHM EmergencyMessageDelivery(Source, Destination, Message)
INPUT: Source, Destination, message priority, current route state
OUTPUT: Delivered, buffered or failed status
BEGIN
IF Destination is directly reachable THEN
SEND Message to Destination
ELSE IF a valid AODV route exists THEN
FORWARD Message along the route
ELSE
Initiate AODV route discovery by broadcasting RREQ
WAIT for RREP until discovery timeout
IF RREP received THEN

Install route and FORWARD Message
ELSE
Store Message with timestamp and priority
Retry discovery at the next permitted interval
END IF
END IF
IF an active link fails THEN
Mark route INVALID and send RERR to affected source
Use alternate valid route if available
END IF
IF a Mobile IP registration is required and a mobility agent is reachable THEN
Obtain CoA and register with HA
HA updates binding and tunnels packets to CoA
END IF
Record delay, delivery status and control-message count
RETURN final status
END

Stage 4 – Performance evaluation plan

The following are example calculations using assumed values to show how results should be reported. They are not actual simulator measurements; replace them with measured results after running the test.

Metric	**Formula**	**Illustrative test result**
Packet delivery ratio	Packets received ÷ packets sent × 100	180 ÷ 200 × 100 = 90%
Packet loss ratio	(Sent − received) ÷ sent × 100	(200 − 180) ÷ 200 × 100 = 10%
Average end-to-end delay	Sum of delivery delays ÷ delivered packets	3.6 s ÷ 180 = 0.020 s = 20 ms

Routing overhead	Routing control packets transmitted	Example: 75 control packets for the test
Route recovery time	Time from link failure until a valid route is restored	Measure with simulation timestamps

Test plan

Test case	**Change**	**Record**
Test 1	10 mobile nodes, low mobility	Delay, delivery ratio, control packets
Test 2	20 mobile nodes, medium mobility	Delay, delivery ratio, control packets
Test 3	30 mobile nodes, high mobility	Delay, delivery ratio, control packets, route failures
Test 4	Disable one relay/link	Route recovery time, buffered and delivered messages
Test 5	Temporarily remove infrastructure gateway	Confirm local ad hoc communication and buffering behaviour

Stage 5 – Cross-defence answers

Why is AODV appropriate? AODV discovers routes when needed and reacts to active-link failure. This can reduce periodic update traffic compared with proactive routing when communication is intermittent. It may, however, incur discovery delay and repeated control traffic in very high mobility.

How does Mobile IP maintain communication during movement? The MN obtains a new CoA in a visited network and registers it with the HA. The HA updates the home-address binding and tunnels packets addressed to the home address to the current CoA. This requires reachable mobility infrastructure and a valid registration.

Stage 6 – Joint evaluation and conclusion

- The ad hoc mesh can be deployed quickly and can continue local communication without fixed base stations, provided usable wireless links remain.
- Mobile IP is useful when a mobile endpoint must retain a stable home-address identity across supported networks; it does not replace ad hoc routing.
- AODV is suitable for on-demand route discovery in a changing topology, but high mobility can increase discovery delay and control overhead.
- Critical messages should be prioritised, authenticated and buffered during temporary outages.

- If all communication links are unavailable, immediate delivery is impossible; the system should store messages and retry when connectivity returns.
- Before deployment, evaluate node density, radio range, mobility speed, bandwidth, energy, security, latency limits and the availability of gateways.

Proposed network layers

- The local communication layer consists of ambulances, police vehicles, rescue teams and temporary relay nodes. These devices form an ad hoc wireless mesh so that nearby teams can exchange information without waiting for a fixed base station.
- The routing layer uses AODV to discover paths when communication is required and to report broken routes. If a usable alternate path exists, traffic can be redirected; otherwise, the application stores messages for later delivery.
- An optional gateway connects the local mesh to a surviving cellular, satellite or fixed network. Mobile IP can support stable addressing for devices communicating across supported home and visited networks, but it does not replace the ad hoc routing function.

Message priority and failure handling

- Emergency alerts, rescue requests and patient-critical information should receive higher priority than routine status reports. The queue should preserve message identifiers and timestamps so that retries do not create confusing duplicates.
- If a route fails, AODV route maintenance should invalidate the failed path and notify affected sources. The source can attempt an alternate route or initiate discovery again when the network is available.
- If all links are unavailable, immediate delivery is not possible. The system should retain the message, record its pending status and retry when connectivity returns rather than silently discarding critical information.

Experimental test cases

- Test a small network with a few mobile nodes, then increase the number of ambulances, rescue teams and relays. This reveals how contention and routing-control traffic change as the network grows.
- Compare low, medium and high mobility. Higher movement speeds are expected to cause more link changes and route repairs, but the actual effect must be measured in the selected simulator or test environment.
- Include link-failure tests in which a relay node moves out of range or loses power. Record whether an alternate route is found, how long recovery takes and how many application packets are lost.

Performance indicators and calculations

- Packet delivery ratio is calculated as received data packets divided by sent data packets, multiplied by 100. Packet loss percentage is lost packets divided by sent packets, multiplied by 100.
- Average end-to-end delay is the sum of delivery delays for received packets divided by the number of received packets. Routing overhead can be reported as the number of control packets or control bytes transmitted during the test.

- Results should include the simulation duration, node count, mobility speed, traffic rate, packet size and random seed. The numerical values already shown in the worked example are illustrative only and must not be presented as measured results.

Security, limitations and deployment considerations

- Nodes should authenticate one another before accepting critical routing or mobility information. Encryption, access control and secure key management protect operational and patient information.
- Ad hoc networks can be affected by interference, congestion, limited battery capacity and malicious or malfunctioning nodes. Redundant relays and periodic health checks can improve resilience but add equipment and management cost.
- Before deployment, planners should verify radio coverage, available spectrum, expected node density, power endurance, message priorities, gateway availability and the maximum acceptable delay for each message type.

Overall engineering justification

- The selected design separates three responsibilities: ad hoc routing provides local packet paths, Mobile IP maintains stable addressing where mobility infrastructure is available, and the application layer prioritises and stores messages.
- This separation avoids treating Mobile IP as a complete routing solution or assuming that AODV can guarantee connectivity when no wireless path exists. Each mechanism addresses a different part of the communication problem.
- The final design is appropriate as a resilient emergency communication proposal, subject to measured validation. Deployment decisions should be based on repeatable experiments and field tests under realistic mobility and failure conditions.

Final conclusion: A practical emergency network combines an ad hoc multi-hop layer, AODV route discovery and repair, optional Mobile IP for supported external connectivity, and buffering for disconnected periods. The design must be tested under different node counts and mobility conditions before its performance is claimed.

Submission note

The numerical values in the worked examples are illustrative assumptions used to demonstrate calculations. Run the required simulation or experiment and replace these values with the actual measured results before submission.